

A Brain Network Exploration Through Random Network Models

University of Calgary

Department of Physics and Astronomy

Advanced Data Analysis (PHYS 605)

Course Instructor: Dr. Claudia Gomes Da Rocha

Final Project by

B.S. Claudeth Clarissa Hernández Alvarez

ABSTRACT

This project investigates the intricate organization of the human brain through network analysis. A brain network, derived from the Connectivity-based Brain Imaging Database at BNU, is employed, where each edge in our network data represents a fiber tract. A sample of 9119 nodes is thoroughly explored at the network level. For comparison, an Erdos-Renyi random network, matched in size and average degree, is constructed. The evaluation encompasses average clustering coefficient and degree distribution. The dataset is successfully fitted with a power-law model, and our networks exhibit a small-world property.

Introduction & Background

1.1 The Brain as Network Data

The brain is one of the most complex structures that exist, housing around trillions of neurons with trillions of synapses among them. A deeper study of the brain could be highly beneficial for understanding it, leading to advances in the study of various neurological diseases. When it comes to examining complex structures, one of the most efficient mathematical tools is graph theory, which is employed in network analysis for such studies [1].

The aim of this work is to conduct a network level statistical analysis of the brain. The network we will be working with is presented in form of an edges list, where each edge represents unweighted neural fiber tracts. These data originate from the *Connectivity-based Brain Imaging Research Database* (C-BIRD) at Beijing Normal University (BNU) [2]. The data were obtained from one of 57 volunteers who completed two MRI scan sessions with an interval of approximately 6 weeks. They were right-handed and had no history of neurological or psychiatric disorders. The dataset was acquired by the Network Data Repository [3] and processed by a turn-key pipeline which used structural and diffusion MRI data to estimate multi-resolution connectomes reliable and scalable, through the MRI Cloud of the *NeuroData* repository [4].

1.2 Network Science as a tool

A graph consists of a mathematical element consisting of nodes and edges (see a) in Figure 1). There are many types of graphs, and these can be classified according to their shape, size, density, or other properties that we will explore later. Network science is a comprehensive study that combines graph theory with the science of complexity, encompassing the study of real networks, from social networks and the World Wide Web to biological networks and the brain's connectome [5, 6].

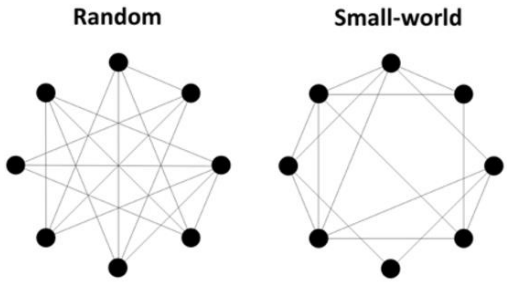


Fig 1. *Two known types of Graphs*

The properties most studied in a network include its degree distribution, clustering, diameter, modularity, robustness, etc. In this study, we will generate a random *Erdos-Renyi network* and compare it with the previously described brain network. In both cases, we will primarily examine the degree distribution and clustering in detail, as well as a well-known property often observed in real networks, such as those mentioned earlier: the *small-world* property.

2.1 Random Network vs Real Network properties

The first step involved creating a *Erdos-Renyi* network, where links are generated based on the number of nodes, *N*, and probability *p*. The properties of these random networks are well-known. In our code, we visually confirmed key aspects, such as, that the probability mass function (PMF) of node degrees follows a binomial distribution. Also, that when the average degree is significantly smaller than the number of nodes, the PMF tends toward a Poisson distribution. It was also graphically validated that the clustering coefficient (CC) of each node, and its average is approximately equal to the probability with which the links were assigned [7].

Subsequently, the BNU dataset was imported, and the average degree and CC's were obtained. The equivalent random network was generated assuming that the link probability, *p*, is given by,

$$p = \frac{\langle k \rangle}{N - 1} \ , \tag{1}$$

where *N* is the number of nodes and $\langle k \rangle$ is the average degree of the Network.

In the same way as was done with the random network, a graph was obtained to visualize the PMF of the degree (both in linear and log-log scales) of our network. This allowed us to compare it with the Poisson distribution of the equivalent random network. Additionally, the CC of each node was calculated and graphed. Along with the average CC, it was compared to the value *p* (representing the CC value of the equivalent random network).

2.2 Network Model Analysis of the Brain Network

Continuing with the analysis of the degree distribution, the PMF was changed into a Probability Density Function (PDF), by changing the spacing of the histogram into logarithmic bins to obtain an equidistant separation according to the logarithmic scale.

Many real networks, such as social, computer, biological, and ecological networks, exhibit a scale-free behavior. This means that the distribution of the degree, *k*, unlike a random network, follows a power-law (see equation 2) [5]. Therefore, next, we will explore how we can find an effective power-law model that fits our data.

$$P \propto k^{-\alpha} \tag{2}$$

Fortunately, Clauset, Shalizi, and Newman [8] developed a methodology to determine "the goodness-of-fit" of the data distribution to a power-law and how we can estimate the parameters of our fit once it has been confirmed whether it follows a power-law or not.

We will start by estimating the parameters necessary for our fit. The key value we need to calculate is α. Starting from,

$$P(k) = \frac{\alpha - 1}{k_{min}} \left(\frac{k}{k_{min}} \right)^{-\alpha}. \quad (3)$$

It should be clarified that not the entire domain of the distribution will behave like a power law; we will fit the model starting from a value, which we will call, k_{min}, which is the same as that found in Equation 3 [8]. The way we will calculate the coefficient α is through the maximization of the natural logarithm of the likelihood [7,8], so that the α coefficient that maximizes the likelihood is,

$$\hat{\alpha} = 1 + N \left[\sum_{i=1}^{N} ln \frac{k}{k_{min}} \right]^{-1}.$$

In our code, we carry out this procedure using the Power-Law library of Python.

Up to now, we have stated that the value of k_{min}, is the one that determines from where we will start fitting the power-law, but what kind of fit can we give before that value? Clauset et al. propose that we can have a good overall fit of the model if the values before k_{min}, follow an exponential distribution. The expression for the fitting function would be,

$$P(k) = \begin{cases} C \left(\frac{k}{k_{min}} \right)^{-\alpha} & for \quad k \geq k_{min} \\ C e^{-\alpha \left(\frac{k}{k_{min}} - 1 \right)} & for \quad k < k_{min} \end{cases}$$

where *C* is the normalization constant. Due to this dual behavior, a test will be conducted to see if the power-law is a plausible fit for our data using a *p*-value, also comparing with an exponential function. The statistical test we will perform is a *Kolmogorov-Smirnov* (KS) test. The closer the *p*-value is to 1, the better the fit to a power-law [8].

3.1 Random Network vs Real Network properties

In Figure 2, we are comparing a Poisson distribution with our brain network, and it is evident that our brain network significantly differs from its equivalent random network. While the random network is expected to follow a Poisson distribution, ours appears to exhibit scale-free behavior. One of the notable signs of this behavior is the presence of a long tail, indicating that very few nodes have a significantly higher number of connections, orders of magnitude larger than most nodes in the network that follow a smaller distribution.

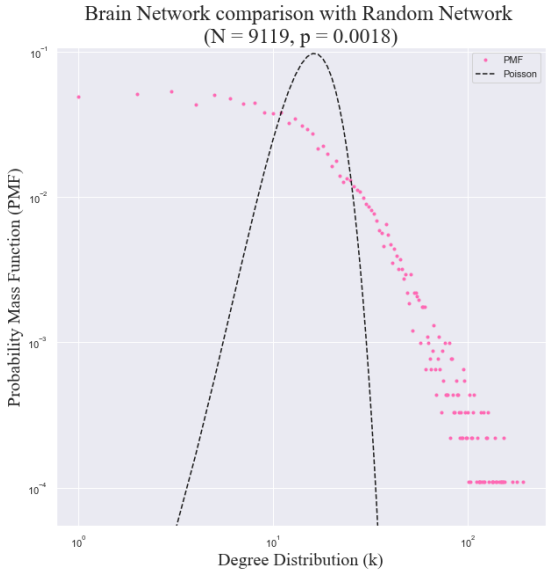


Fig 2. *The PMF of degree of a random network vs the PMF of our Brain Network.*

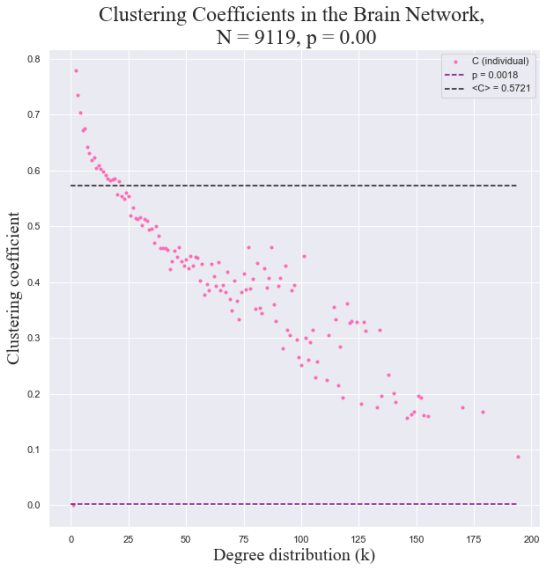


Fig 3. *The Clustering coefficients vs the degree distribution of the Brain Network.*

In Figure 3, we depict the clustering coefficient as a function of the degree distribution in our brain network. It is noticeable that as the node degree increases, the clustering coefficient tends to decrease. This trend suggests that nodes with higher degrees tend to have a lower clustering coefficient, indicating a reduction in the local interconnectedness among their neighbors. In simpler terms, the decrease in clustering coefficient with increasing degree may be indicative of a more distributed or less clustered network structure compared to nodes with lower degrees.

Moreover, it is noteworthy that the values of link probability and average clustering coefficient differ significantly in this case.

3.2 Network Model Analysis of the Brain Network

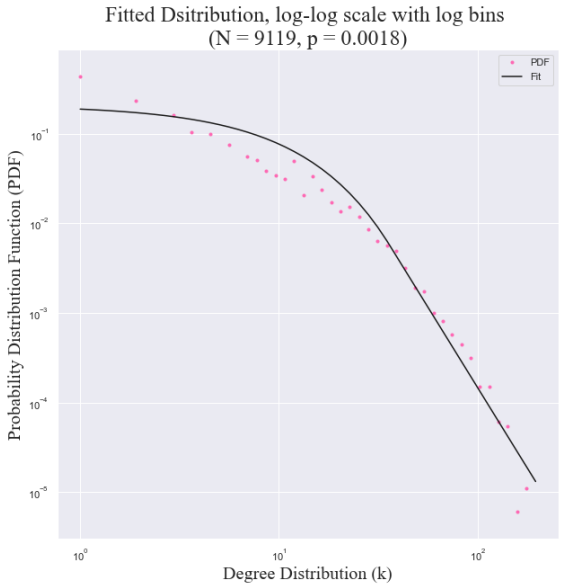


Fig 3. *The PDF of the degree in our Network with it respective fitting*

Now, in Figure 3, we transformed the Probability Mass Function (PMF) into a Probability Density Function (PDF) by spacing the bins. This resulted in a much less noisy and smoother figure, an elongation of the y-axis scale, and the removal of tails. Subsequently, we performed a likelihood maximization fit that aligns quite well with our data distribution. The fitting process was optimized to maximize the likelihood of the observed data under the proposed model. The successful alignment between the fitted model and our actual data points demonstrates the effectiveness of the selected model in capturing the underlying patterns of the observed network, providing a refined and accurate representation.

Evidently, it appears to exhibit a scale-free behavior. Subsequently, we assessed the goodness-of-fit of our degree distribution data using a power-law model compared to an exponential model. The power-law fit was performed using the *Power law* library. The *Kolmogorov-Smirnov* test resulted in a statistic of approximately 2.04 and a p-value of approximately 0.81. The high p-value suggests that there is insufficient evidence to reject the null hypothesis, indicating that our degree distribution is consistent with a power-law distribution rather than an exponential distribution.

Conclusion

This project delves into the organization of the human brain through network analysis. Utilizing a brain network derived from the Connectivity-based Brain Imaging Database at BNU, each edge in our network represents a fiber tract. Our exploration covers a comprehensive analysis of a sample comprising 9119 nodes at the network level. For comparative purposes, we construct an Erdos-Renyi random network, matching the size and average degree. The assessment includes parameters such as average clustering coefficient and degree distribution. Notably, our dataset is adeptly fitted with a power-law model, signifying a scale-free network structure. Additionally, our networks exhibit a small-world property, highlighting efficient information transfer with a balance of local clustering and global connectivity. This investigation provides valuable insights into the underlying network properties of the human brain, shedding light on its complex and organized nature.

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